

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Avinash K	Reg. No.:	192411044
Section / Batch:	CSE	Date:	

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1. Large-Scale Image Merge System

An image processing system recursively divides an image dataset into two halves but performs quadratic work at each level because of inefficient merging operations. Apply the substitution method to solve the recurrence $T(n)=T(n/2)+n$. Expand the recurrence step-by-step and derive the final complexity. Analyze the growth rate for large image datasets. Interpret the result using asymptotic notation.

Q2. Distributed Sensor Network Analysis

A sensor network algorithm divides a monitoring problem into three equal subproblems and performs only constant work at each level for control operations. Apply the substitution method to solve the recurrence $T(n)=3T(n/3)+1$. Show all intermediate steps and derive the final asymptotic complexity. Analyze how this algorithm scales with increasing sensor nodes. Interpret the efficiency using asymptotic notation.

Code of Conduct

I Avinash k / 192411044 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Avinash k

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Q13. Large scale Image Merge System

Given Recurrence

$$T(n) = T(n/2) + n$$

$$\text{assume, } T(1) = 1$$

S1:

$$T(n) = T(n/2) + n$$

$$T(n) = [T(n/4) + n/2] + n = T(n/4) + 3n/2$$

S2:

$$T(n) = [T(n/8) + n/4] + 3n/2 = T(n/8) + 7n/4$$

S3:

$$T(n) = T(n/2^k) + n(1 + 1/2 + 1/4 + \dots + 1/2^{k-1})$$

S4:

$$1 + 1/2 + 1/4 + \dots + 1/2^{k+1} = \frac{1 - (1/2)^{k+1}}{1 - 1/2} = 2(1 - 1/2^{k+1})$$

$$T(n) = T(n/2^k) + 2n(1 - 1/2^k)$$

S5:

$$n/2^k = 1$$

$$2^k = n$$

$$k = \log_2 n$$

Substitute

$$T(n) = T(1) + 2n(1 - 1/n) = 1 + 2n - 2 = 2n - 1$$

Final complexity:

$$T(n) = n - 1 \quad T(n) = O(n)$$

Grate: algorithm grows linearly with input size

$$\bullet \text{ Big-O: } O(n) \quad \text{Big-}\Omega: \Omega(n) \quad \text{Big-}\Theta: \Theta(n)$$

Q.14 Distributed Sensor Network Analysis

Recurrence:

$$T(n) = 3T(n/3) + 1$$

$$T(1) = 1$$

S1: $T(n) = 3T(n/3) + 1 = 3[3T(n/9) + 1] + 1 = 9T(n/9) + 4$

S2: $T(n/9)$ expand

$$= 9[3T(n/27) + 1] + 4 = 27T(n/27) + 13$$

S3:

k expansions

$$T(n) = 3^k T(n/3^k) + (1 + 3 + 3^2 + \dots + 3^{k-1})$$

S4:

sum of Geometric series

$$1 + 3 + 9 + \dots + 3^{k-1}$$

$$= \frac{3^k - 1}{3 - 1} = \frac{3^k - 1}{2}$$

$$T(n) = 3^k T(n/3^k) + \frac{3^k - 1}{2}$$

S5: Reach the Base Case

$$n/3^k = 1 \quad 3^k = n \quad k = \log_3 n$$

$$T(n) = nT(1) + n^{1/2} = n + n^{1/2} = 3n^{1/2}$$

final complexity

$$T(n) = 3n^{1/2}$$

$$T(n) = \Theta(n)$$

Growth rate:

~ The algorithm scales efficiently for large sensor networks

notation

$$\text{Big-O: } O(n)$$

$$\text{Big-}\Omega: \Omega(n)$$

$$\text{Big-}\Theta: \Theta(n)$$